MIL-L-43283B
16 June 1981
SUPERSEDING
MIL-L-43283A
6 October 1969

MILITARY SPECIFICATION

*LEATHER, CALFSKIN, KIP, AND CATTLEHIDE FOR

FOOTWEAR UPPERS, CHROME TANNED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers smooth finish, full grain calfskin and burnished full grain, calfskin, kipskin, and cattlehide upper leather.

* 1.2 Classification. The leather shall be furnished in the following types as specified (see 6.2):

Type I - Calfskin
Type II - Kip
Type III - Cattlehide sides

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-L-43432 - Leather, Packaging and Packing Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be address to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8330

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STANDARD

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Material. The leather shall be made from brined or green-salted calfskin, kipskin or cattlehide, as specified (see 1.2).

3.1.1 Type I. Type I shall be full grain calfskin or burnished full grain calfskin.

3.1.2 Type II. Type II leather shall be burnished full grain kip.

* 3.1.3 Type III. Type III leather shall be burnished full grain cattlehide.

3.2 Tannage. The leather shall be chrome tanned.

3.3 Physical requirements

* 3.3.1 Trim (form and area). Type I leather shall be furnished in the form of whole skins. Type II leather shall be furnished in the form of sides or whole skins. The type III leather shall be furnished in the form of sides. The leather shall conform with the applicable definitions listed under "Definition Of Terms" in FED-STD-311. The type I calfskin shall not exceed 20 square feet in area. The type II side shall not exceed 17-3/4 square feet and the whole kipskin shall not exceed 35-1/2 square feet in area. Type III cattlehide shall not exceed 22 square feet in area. The area in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.3.2 Color. The color of the leather shall be as specified (see 6.2). The leather shall be dyed on the grain side and flesh side. The dye shall penetrate both sides of the leather, but need not be struck entirely through the leather. Leather previously dyed or finished in a color not specified and then redyed or refinished to match a specified color shall not be acceptable.

3.3.3 Finish. The leather shall have a smooth, bright, pigmented finish on the grain side. Embossed or printed leather shall not be acceptable. A lacquer emulsion top finish based essentially on nitrocellulose compounding shall be applied to the grain side of all leather except glazed calfskin.

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3.3.3.1 Burnishing. Burnishing shall be accomplished with automatic machines equipped with 400 grit or 320 dull grit cloth or paper. The burnishing operation shall slightly level out the peak strata of the grain, but shall not remove all evidence of the grain. Snuffing or buffing to correct surface blemishes shall not be permitted.

* 3.3.4 Thickness. The leather shall be furnished in the following thickness.

Type	Thickness (ounces)
I	2 to 4
II	3-1/4 to 4-1/2
III	3-1/4 to 4-1/2

NOTE: 1 ounce = 1/64 inch

3.3.5 Stitch tearing strength. Not less than 80 percent of the specimens tested shall have a minimum stitch tearing strength of 30 pounds when tested as specified in 4.2.2.2.

3.3.6 Shrinkage temperature. The leather shall have a minimum shrinkage temperature of 97°C when tested as specified in 4.2.2.2.

3.4 Chemical requirements.

3.4.1 Leather. The leather shall conform to the chemical requirements of table I when tested as specified in 4.2.2.2.

TABLE I. Chemical requirements

Characteristic	Minimum	Maximum
Chloroform - soluble material, percent 1/	3.0	12.0
Chromic oxide, percent 1/	3.0	7.0
pH value	3.0	4.5

1/ Calculated on the moisture free basis.

3.4.2 Paranitrophenol. Unless otherwise specified (see 6.2), the leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.2.2.2.

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3.5 Workmanship. The leather shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the appplicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with FED-STD-311.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection.

* 4.2.2.1 End item examination. The leather shall be examined for the visual and dimensional defects indicated in 4.2.2.1.1 and 4.2.2.1.2. The sample unit for these examinations shall be one side, or skin. The lot size shall be expressed in units of one side or skin. Sampling for these examinations shall be performed in accordance with FED-STD-311 except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more sides or skins. Separate examinations shall be performed for visual and dimensional characteristics, but the same sample may be used for each examination.

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*4.2.2.1.1 Visual examination.

Examine	Defect
Material	
Type I	Not full grain or burnished full grain calfskin.
Type II	Not burnished full grain kip.
Type III	Not burnished full grain cattlehide.
Trim	
Type I	Not whole skins.
Type II	Not whole skins or sides.
Type III	Not sides.
Color	Not as specified or not uniform. Dye does not penetrate grain and flesh side. Evidence that leather has been redyed or refinished to cover unspecified color.
Finish	Finish not smooth. Finish not bright. Finish peeled, chipped or tacky. Grain snuffed or buffed. Finish missing.
Quality of leather	Cut, hole, open or badly healed scratch, tear, stain, deep fat wrinkle, heavy vein, hip mark, rough shoulder, coarse grain, pipey area, loose flesh or other surface imperfection which individually or collectively impair the cutting value of more than 10 percent of the area of the leather.

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* 4.2.2.1.2 Dimensional examination.

Examine	Defect
Area (expressed in terms of square feet)	Less than area marked on leather.
Type I	Area exceeds 20 square feet.
Type II	Area of kip side exceeds 17-3/4 square feet. Area of whole skin exceeds 35-1/2 square feet.
Type III	Area of side exceeds 22 square feet.
Marking	Omitted, illegible, or not in area specified.
Thickness 1/	
Type I	Less than 2 ounces or greater than 4 ounces.
Type II	Less than 3-1/4 ounces or greater than 4-1/2 ounces.
Type III	Less than 3-1/4 ounces or greater than 4-1/2 ounces.

1/ Thickness shall be determined in accordance with Method 1011 of FED-STD-311. No individual thickness value shall be less than the minimum or greater than the maximum specified.

4.2.2.2 End item testing. The methods of testing specified in FED-STD-311 wherever applicable and as listed in table II, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exist:

a. More than three test failures occur for stitch tearing strength.

b. More than one test failure occurs for shrinkage temperature.

c. Any composite test result fails to meet the specified requirement.

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TABLE II. End item testing

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Material identification	3.1	1/	-	-
Tannage	3.2	1/	-	-
Finish	3.3.3	1/	-	-
Stitch tearing strength	3.3.5	2151	X	-
Shrinkage temperature	3.3.6	7011	X	-
Chloroform-soluble material	3.4.1	6341	-	X
Chromic oxide	3.4.1	6515	-	X
pH value	3.4.1	6621	-	X
Paranitrophenol	3.4.2	6711	-	X

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.2.3 Packaging inspection. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that preservation packaging, packing and marking comply with the section 5 requirements.

5. PACKAGING

* 5.1 Preservation packaging. Preservation packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be preserved packaged in accordance with the applicable requirements of MIL-L-43432.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. The leather is intended for use as upper leather in Department of Defense dress shoes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Color required (see 3.3.2).
(d) When paranitrophenol fungicide is not required (see 3.4.2).
(e) Selection of applicable levels of preservation packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-L-43283B Leather, Calfskin, Kip, And Cattlehide For Footwear Uppers, Chrome Tanned

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